

UNIT I: FOUNDATIONS OF ARTIFICIAL INTELLIGENCE & INTELLIGENT AGENTS

Weightage: 25% | Syllabus Section 1

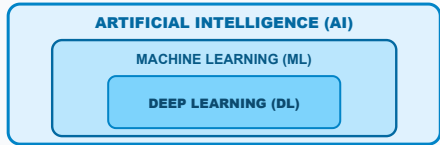
1. AI Definitions, Turing Test & Ethics

AI Definition: Computational systems capable of performing tasks requiring human intelligence (perceiving, reasoning, learning, acting).

Turing Test (1950): Interrogator C chats blind with Human A and Machine B. Machine passes if C cannot reliably differentiate machine from human (>30% indistinguishability after 5 min).

Ethics & Bias: Algorithmic fairness, privacy preservation (data theft), transparency/explainability (XAI), job displacement, autonomous lethal weapons.

2. AI vs. ML vs. DL Hierarchy

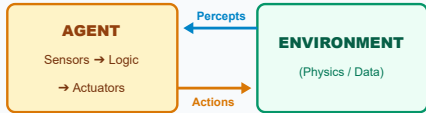


AI: Broadest science of intelligent machines.

ML: Statistical algorithms learning patterns from data without explicit rule programming.

DL: Multi-layer Artificial Neural Networks (ANNs) for automatic feature extraction.

3. Intelligent Agents & PEAS Framework



PEAS: Performance Measure, Environment, Actuators, Sensors. (e.g. Self-Driving Car: Safety/speed, Roads/traffic, Steering/brakes, Cameras/LiDAR).

Agent Class	Internal Working Principle	Decision Mechanism	Canonical Real-World Example
Simple Reflex Agent	Acts solely on current percept via Condition-Action rules (ignores history).	IF condition THEN action lookup.	Thermostat (IF Temp < 20°C THEN Turn Heater ON).
Model-Based Reflex	Maintains internal state reflecting unobserved parts of current world.	State + Rules → Next Action selection.	Automated Vacuum Cleaner tracking cleaned vs dirty rooms.
Goal-Based Agent	Combines internal state with explicit Goal definitions to plan actions.	Search & Planning algorithms (A*, BFS).	Autonomous GPS Navigation System finding destination route.
Utility-Based Agent	Evaluates multiple goals using a continuous Utility Function U(s).	Maximizes expected trade-off utility.	Algorithmic High-Frequency Trading Bot balancing profit vs risk.
Learning Agent	Separated into: Learning Element, Critic, Learning Goals, Performance Element.	Modifies actions based on feedback.	Self-driving car improving braking smoothly through experience.

UNIT II: SEARCHING ALGORITHMS & KNOWLEDGE REPRESENTATION

Weightage: 25% | Syllabus Section 2

1. State Space Search: Uninformed vs. Heuristic

Algorithm	Frontier Data Structure	Time / Space	Complete?	Optimal?
BFS	FIFO Queue	$O(b^d)$ / $O(b^d)$	Yes (if b finite)	Yes (unit cost)
DFS	LIFO Stack	$O(b^m)$ / $O(b \cdot m)$	No (infinite depth)	No
Hill Climbing	Current State only	$O(m)$ / $O(1)$	No (local extrema)	No
A* Search	Priority Queue (min f)	$O(b^d)$ / $O(b^d)$	Yes	Yes (if h admissible)

A* Evaluation Function: $f(n) = g(n) + h(n)$

- $g(n)$ = Exact cost from start node to node n.
- $h(n)$ = Heuristic estimated cost from n to goal.
- Admissibility Condition:** $0 \leq h(n) \leq h^*(n)$ (Never overestimates true cost).
- Data-Driven vs Goal-Driven:** Forward chaining (data → goals) vs Backward chaining (hypothesis → facts).

2. Knowledge Representation & Resolution Refutation

KR Techniques: Propositional Logic (Booleans), First-Order Predicate Calculus (FOPC: $\forall, \exists$, predicates, functions), Semantic Networks (graph of nodes & is-a/has-a edges), Frames (slots & fillers with default inheritance), Scripts (stereotypical event sequences).

8-Step CNF Conversion Pipeline for FOPC:

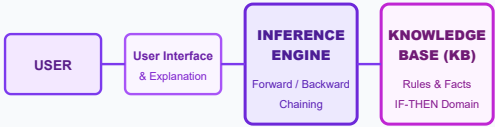
- Eliminate $\Leftrightarrow \equiv (A \Rightarrow B) \wedge (B \Rightarrow A)$
- Eliminate $\Rightarrow \equiv \neg A \vee B$
- Push Negations Inward (De Morgan's & $\neg \forall x P(x) \equiv \exists x \neg P(x)$)
- Standardize variables (rename overlapping variables)
- Skolemization:** Drop $\exists$ by substituting constants or Skolem functions $f(x)$
- Drop Universal Quantifiers ($\forall$)
- Distribute $\vee$ over $\wedge$ to form Conjunction of Disjunctions
- Separate into discrete clauses.

Resolution Refutation Algorithm: To prove Query Q: (1) Negate Q ($\neg Q$), (2) Add $\neg Q$ to KB in CNF, (3) Resolve complementary literals ($A \vee B$) and ($\neg B \vee C$) → ($A \vee C$) until **Empty Clause** ($\square$) is derived ⇒ Contradiction proves Q!

UNIT III: EXPERT SYSTEMS & KNOWLEDGE ENGINEERING

Weightage: 25% | Syllabus Section 3

1. Expert System Architecture & Components



Core Components:

- **Knowledge Base:** Declarative facts & heuristic IF-THEN rules.
- **Inference Engine:** Brain of ES executing deduction via pattern matching (Rete algorithm).
- **Explanation Facility:** Answers "HOW" a conclusion was reached and "WHY" an input is asked.
- **Working Memory:** Dynamic state tracking current case facts.

2. Knowledge Engineering & Case Studies

Knowledge Engineering Lifecycle:

1. **Knowledge Acquisition:** Eliciting expertise from Domain Experts via interviews, protocols, and observation.
2. **Knowledge Representation:** Formalizing raw knowledge into rules, frames, or ontologies.
3. **Knowledge Validation & Verification:** Checking consistency, completeness, and absence of circular rules.
4. **Inference Deployment:** Forward Chaining (data-driven: starts with symptoms → fires rules → infers disease) vs Backward Chaining (goal-driven: starts with suspected disease → checks if symptoms match).

Benchmark Expert Systems:

- **MYCIN (1970s):** Diagnosed bacterial blood infections (meningitis) using certainty factors (CF).
- **DENDRAL:** Inferred chemical molecular structures from mass spectrometer data.

UNIT IV: INTRODUCTION TO MACHINE LEARNING ALGORITHMS & METRICS

Weightage: 25% | Syllabus Section 4

1. Supervised Learning

Goal: Learn mapping $y = f(x)$ from labeled dataset $\{(x_i, y_i)\}$.

Linear Regression: Fits continuous output line $y = \theta_0 + \theta_1 x$ minimizing Mean Squared Error: $J(\theta) = \frac{1}{2m} \sum (h_{\theta}(x_i) - y_i)^2$.

Decision Trees: Recursive binary partitioning using Information Gain: $\text{Gain}(S, A) = \text{Entropy}(S) - \sum (|S_v|/|S|) \text{Entropy}(S_v)$, where $\text{Entropy}(S) = - \sum p_i \log_2(p_i)$.

Naive Bayes: $P(y|X) = \frac{P(X|y) \cdot P(y)}{p}$ (X) under conditional independence assumption.

k-Nearest Neighbors (k-NN): Lazy, non-parametric instance classifier using Euclidean metric: $d(p, q) = \sqrt{(\sum (p_i - q_i)^2)}$.

2. Unsupervised Learning

Goal: Discover hidden structure/patterns in unlabeled data $\{x_i\}$.

k-Means Clustering:

1. Select k random centroids $\mu_1, \dots, \mu_k$.
2. Assign each point to nearest centroid: $c_i = \text{argmin}_j ||x_i - \mu_j||^2$.
3. Update centroids as cluster means: $\mu_j = (1/|C_j|) \sum_{\{i \in C_j\}} x_i$.
4. Repeat until centroids converge (no shift).

Apriori Association Mining:

- **Support:** $\text{supp}(X \rightarrow Y) = P(X \cup Y)$.
- **Confidence:** $\text{conf}(X \rightarrow Y) = P(Y|X) = \text{supp}(X \cup Y) / \text{supp}(X)$.
- **Lift:** $\text{lift}(X \rightarrow Y) = \text{conf} / \text{supp}(Y)$.
- **Apriori Principle:** Any subset of a frequent itemset must also be frequent.

3. Confusion Matrix & Evaluation Metrics

Metric Formula	Core Purpose & Exam Context
Accuracy = $\frac{TP+TN}{TP+TN+FP+FN}$	Overall correctness; misleading in imbalanced datasets (e.g. 99% benign).
Precision = $\frac{TP}{TP + FP}$	Fraction of predicted positives that are true (vital when False Positive is costly: Spam).
Recall (Sensitivity) = $\frac{TP}{TP + FN}$	Fraction of actual positives captured (vital when False Negative is lethal: Cancer).
F1-Score = $2 \cdot \frac{\text{Prec} \cdot \text{Rec}}{\text{Prec} + \text{Rec}}$	Harmonic mean balancing Precision and Recall under class imbalance.

Confusion Matrix:

- **TP:** True Positive | **FP:** Type I Error (False Alarm)
- **FN:** Type II Error (Miss) | **TN:** True Negative

Dimension	Supervised Learning	Unsupervised Learning	Reinforcement Learning (RL)
Training Data	Explicitly labeled inputs + desired ground truth pairs (x_i, y_i) .	Unlabeled raw data points $\{x_i\}$ without target labels.	Dynamic environment feedback (reward/penalty signals r_t).
Feedback Type	Direct supervisory error signal $(y_{pred} - y_{true})$.	No direct error signal; optimizes intrinsic clustering density.	Delayed evaluative scalar rewards through trial-and-error exploration.
Core Task	Classification (discrete class) & Regression (continuous values).	Clustering, Dimensionality Reduction, Association Rules.	Policy optimization, optimal sequential decision making.
Typical Models	Decision Trees, SVM, Naive Bayes, Linear/Logistic Regression, k-NN.	k-Means, Hierarchical Clustering, PCA, Apriori.	Q-Learning, SARSA, Deep Q-Networks (DQN), Policy Gradients.